

Protein Engineering Grammar: From Molecular Formats to Programmable Medicines

Koichiro Saka

Protein therapeutics are conventionally classified by structural format — monoclonal, bispecific, conjugate, fusion. This scheme describes what a molecule is built from, not the functional logic it implements. We present a complementary vocabulary, the Protein Engineering Grammar, that describes an engineered protein as a modular information-processing system. The system comprises four primitives: a Sensor that reads biological or chemical input, a Gate that evaluates a permission condition, a Processor that operates the Actuator, and an Actuator that is the molecule’s effect on external factors. We extend this vocabulary with a Compiler formalism: a seven-stage pipeline from biological problem to molecular architecture. The Compiler separates the choice of which primitive class to deploy from the physical scaffold used to realize that choice. We evaluated this framework against a registry of protein engineering technologies drawn mainly from primary mechanistic literature, all of which fall within its intended scope. The mapping shows that molecules sharing an identical structural classification can implement divergent logic. It also shows that divergent structural scaffolds can implement equivalent logic without extending the core vocabulary. We used the resulting evidence base to identify combinations of primitives that appear absent from the surveyed literature. We evaluated six candidate white-space opportunities along three axes: the number of implementing technologies, market status, and clinical need. One of them emerged as the strongest.

1 Introduction

Protein therapeutics are conventionally classified by structural format: a molecule is a monoclonal antibody, a bispecific antibody, an antibody-drug conjugate, an Fc-fusion protein, or a cytokine fusion. This classification is useful for regulatory, manufacturing, and intellectual-property purposes, but it leaves an important question unanswered. Knowing that a molecule is a bispecific antibody tells a reader what it is built from. It does not tell the reader what functional logic the molecule implements: what condition must be met before it acts, how it combines multiple recognition events, or what physical step converts a recognition event into a therapeutic effect. Two molecules sharing an identical structural classification can differ entirely along this second axis (Figure 1). The reverse also holds: a molecule built from a structural scaffold completely unrelated to antibodies can implement the same logic as an existing antibody technology. A T-cell receptor-based engager [1] recognizes a peptide-MHC complex through a soluble, affinity-matured T-cell receptor domain rather than an antibody paratope. Yet it recruits CD3 to engage a T cell by the same logic as an antibody-based T-cell engager, demonstrating that the underlying Sensor class, Identity, is agnostic to the recognition domain’s structural family: the same functional role can be filled by an antibody Fab, a T-cell receptor, or, as the next example shows, a synthetic peptide macrocycle. A second example makes the same point for the Processor primitive: a fully synthetic, non-immunoglobulin bicyclic peptide conjugate targeting Nectin-4 [2] is built on a chemically constrained macrocyclic peptide of approximately 4 kDa, structurally unrelated to any antibody fragment. Yet it is bound, trafficked, and protease-cleaved to release its payload by the same logic as a conven-

tional antibody-drug conjugate, evidence that the Processor primitive, as defined, captures a pattern more general than the immunoglobulin fold.

We propose a complementary vocabulary, which we call the Protein Engineering Grammar, that describes an engineered protein as a modular information-processing system. It comprises four primitives: a Sensor that reads biological or chemical input, a Gate that evaluates a permission condition, a Processor that operates the Actuator, and an Actuator that is the molecule’s effect on external factors (Section 3). We extend this vocabulary with a Compiler formalism: a seven-stage pipeline from a biological problem, through feature selection and a domain classifier, to a specific molecular architecture and, ultimately, an engineered output. Conventional target-to-drug framing leaves several design decisions implicit: which biological features to sense, what logic they should satisfy, which components realize that logic, and what physical architecture implements that choice. This formalism makes each of them explicit (Section 5).

We built and evaluated this framework against a registry of protein engineering technologies, all of which fall within its intended scope. We drew primarily on primary mechanistic literature (a small number of conference abstracts and company disclosures are flagged as such in Table S1) and applied an explicit evidence hierarchy that distinguishes direct biochemical or structural evidence from cell-based, *in vivo*, author-proposed, or inferred assignments. We used this same evidence base to identify combinations of components that appear absent from the technologies and literature we surveyed (Section 4).

The remainder of this manuscript proceeds as follows. Section 2 situates our approach relative to existing systems-biology and synthetic-biology vocabulary. Section 3 defines the Grammar’s four constituent primitives. Section 4 reports the empirical mapping of these technologies against this vocabulary. This includes cases where a shared structural format conceals divergent logic, cases where divergent scaffolds implement shared logic, and the combinations of primitives that this mapping shows to be absent from the literature we surveyed (white-space). Section 5 introduces the Compiler formalism and its constituent stages, including a Constraint Layer for pharmacokinetic and exposure considerations that sit outside the Grammar’s molecular logic proper. It also briefly touches on Protein Architecture-layer

technology examples that realize a chosen combination of primitives as a manufacturable molecule. Section 6 discusses the implications for AI-driven engineering.

2 Proteins as Information-Processing Systems

An old idea, applied at a narrower grain

Describing a biological molecule as sensing an input, integrating it, and producing an output is not a new idea. Signal transduction biology has long used this vocabulary, and synthetic biology has formalized much of it further, describing engineered genetic circuits explicitly in terms of Boolean logic gates, sense-compute-actuate pipelines, and programmable cellular behavior [3]. Most existing sense-compute-actuate formalisms, moreover, have been developed and validated as engineered genetic circuits operating inside a living cell [3]. Functioning NOT-like gates and related Boolean primitives, for example, are most often demonstrated in engineered-cell contexts that require host-cell genetic modification [4], [5]. What we believe is genuinely underserved is not the vocabulary itself but its systematic application at a specific grain: protein engineering. Our claim differs from this prior work in formalizing the function carried by a single protein molecule.

Relationship to existing frameworks

We compare our approach against the two literatures closest to it, rather than asserting novelty without engaging them.

The dominant organizing scheme for multispecific protein therapeutics remains structural, organizing molecules only by structural class: monoclonal, bispecific, conjugate, fusion. Brinkmann and Kontermann’s widely cited survey [6] catalogs more than 100 distinct bispecific antibody formats, organized by the presence or absence of an Fc region and by symmetric versus asymmetric chain architecture. By design, this scheme answers what a molecule is built from and says little about what logical condition governs its activity. Figure 1 shows four technologies that share a structural classification (a conventional

Why Format-Based Taxonomy Breaks

One conventional 1+1 bispecific format with a shared Direct gate can still implement multiple logic architectures

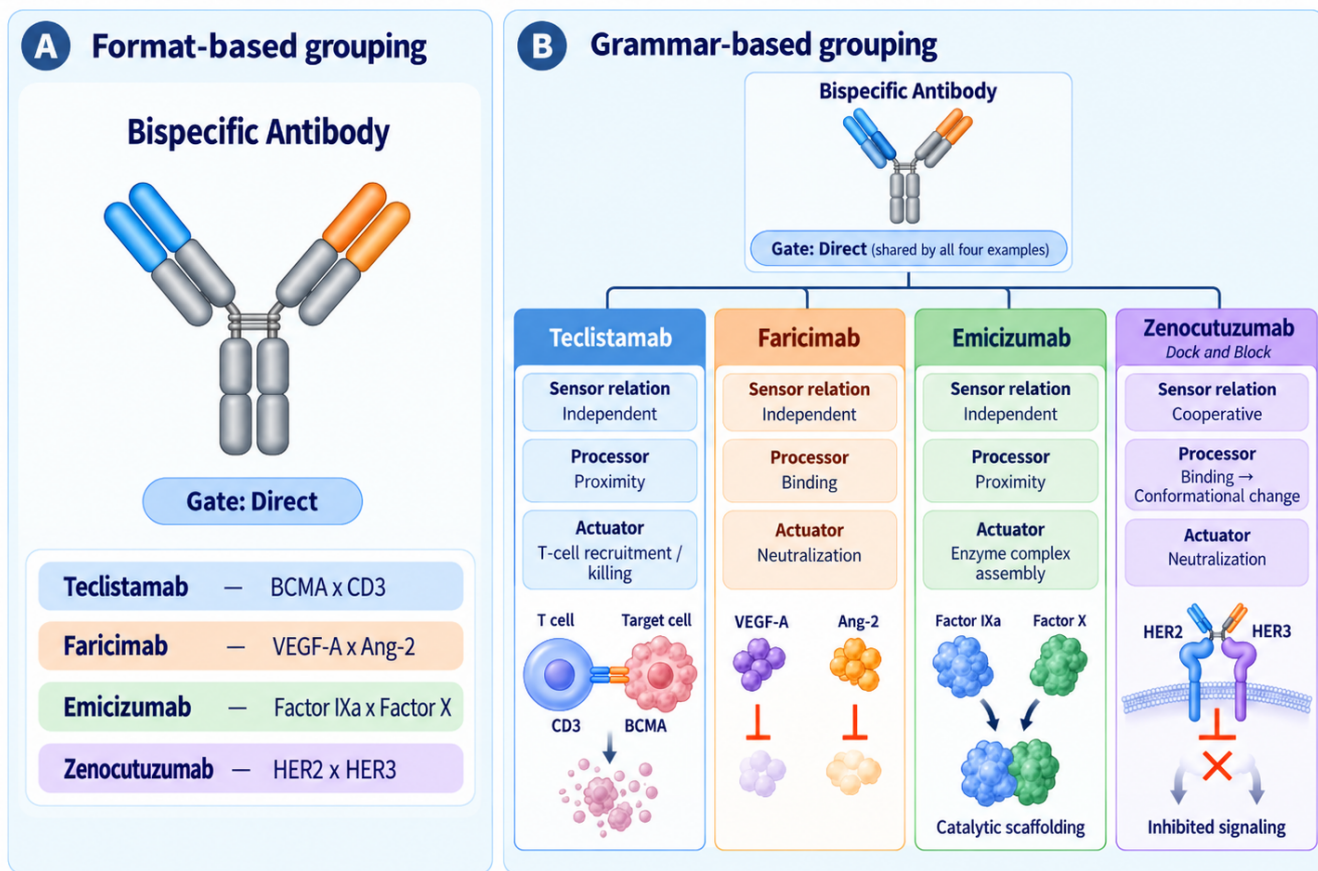


Figure 1: Structural format and functional logic are orthogonal axes. Molecules sharing an identical structural classification, such as “bispecific antibody,” can implement divergent Grammar logic, and molecules built from structurally unrelated scaffolds can implement equivalent logic.

1+1 bispecific IgG) and a Direct Gate but differ in their Sensor relation, Processor, and Actuator; under this scheme, all four would be grouped together as one Fc-containing, IgG-like format. The logic distinctions we report are orthogonal to conventional structure-based classification, not a refinement of it.

A more directly relevant precedent is Labriijn and colleagues’ mechanistic review of the bispecific antibody pipeline [7], which explicitly moves beyond structural classification by distinguishing “obligate” bispecifics — molecules in which physical linkage of two binding specificities creates a functional dependency neither specificity has alone. Labriijn and colleagues further split that dependency into temporal (sequential binding events) and spatial (simultaneous binding events, as in linking an effector cell to a target cell)

categories. This is, to our knowledge, the closest prior mechanistic lens to the one we propose, and we want to state precisely how our contribution differs rather than claim to have invented mechanistic classification outright. First, scope: their categorization is developed for and applied to bispecific antibodies specifically. Our vocabulary, in contrast, is applied uniformly across single-target conditionally activated antibodies, antibody-drug conjugates, cytokine fusions, T-cell receptor-based engagers, and non-immunoglobulin scaffolds — none of which are bispecific in the sense their scheme addresses. Second, granularity: their temporal/spatial distinction is a single axis describing the relationship between two binding events. Our four-primitive vocabulary treats three further questions as separable, each with its own evidence standard: how multiple simultaneous

inputs combine (Sensor’s internal relation axis), what permission condition gates the outcome (Gate), and how the molecule operates the Actuator (Processor). We treat their obligate/temporal/spatial distinction not as a competing scheme, but as one our Sensor-relation and Gate vocabulary can largely recover. We take this agreement between two independently developed frameworks as evidence that the distinction is real, not as a difference to downplay.

On the synthetic-biology side, Brophy and Voigt’s review of genetic circuit design principles [3] formalizes Boolean-logic-inspired transcriptional control in engineered cells. This includes the “tuning knobs” (regulator choice, expression strength) available to circuit designers and the failure modes that arise from operating within a living cell’s constraints. The engineered T-cell logic-gating literature that has grown from this tradition — including AND-gated dual-antigen chimeric antigen receptors [8] and synthetic Notch-based sequential control circuits [5] — implements, to our knowledge, some of the clearest existing examples of protein-level Boolean logic, NOT-like gates included [4]. This Boolean-logic research has, however, been assembled in existing literature chiefly as engineered-cell circuits mediating intracellular-to-extracellular signaling. We extend this vocabulary to a broader class of molecules, including soluble administered proteins that require no genetic modification at all. We do not exclude a technology from the registry solely because it requires genetic modification: the criterion is not the route of expression but the receptor’s own Sensor-through-Actuator logic. The registry accordingly includes three technologies from this tradition as formally mapped examples (Section 4): an orthogonal cytokine-receptor pair, a synthetic Notch (synNotch) receptor, and a de novo protein logic gate that implements NOT-like Boolean logic through the competitive dimer disruption of two designed heterodimeric proteins.

What this framework claims, and what it does not

We therefore treat the Sensor-Gate-Processor-Actuator vocabulary (Section 3) not as a novel invention but as a deliberate transplant and specialization of an existing systems-biology lens onto the domain of protein engineering. To our knowledge, it has not previously been applied systematically there and

mapped against a broad, evidence-graded set of real molecules. The contribution we can substantiate with evidence is the mapping itself: a registry of technologies, most of them with primary-literature support for at least some of their assignments, spanning conventional antibodies, conditionally activated formats, cytokine fusions, T-cell receptor-based engagers, and non-immunoglobulin scaffolds. All are classified against a shared vocabulary with an explicit evidence standard that distinguishes directly demonstrated assignments from inferred ones (Section 4).

3 The Protein Engineering Grammar

A modular vocabulary for engineered proteins

Linguistics uses the term primitive for a grammatical unit — a part of speech, a syntactic category — that does not decompose further. Computer science uses the same term for a basic type or operation that a system does not decompose further either. We propose that an engineered protein — whether a therapeutic antibody, a bispecific engager, a conjugate, or a de novo designed switch — can be described as a modular information-processing system composed of four such primitives. This manuscript addresses the four basic units that constitute the Grammar. A **Sensor** reads biological or chemical input, combining the outputs of more than one simultaneous Sensor instance through an internal relation axis where applicable. A **Gate** evaluates a permission condition on that signal. A **Processor** is the molecule’s internal mechanism that operates the Actuator once the Gate permits it. An **Actuator** is the molecule’s effect on external factors (Figure 2).

This vocabulary is deliberately domain-general. Sensor, Gate, and Processor describe how a protein processes information and passes the result on to the Actuator. These three primitives are treated as core engineering primitives whose classes we attempt to enumerate as exhaustively as the available evidence allows. Actuator, by contrast, denotes the application-specific effect on external factors — what the protein is engineered to accomplish — and the number of possible Actuator classes grows with the number of application domains. We therefore treat Actuator classes as representative rather than exhaustive, drawing our examples from the therapeutic domain while

Protein Engineering Grammar

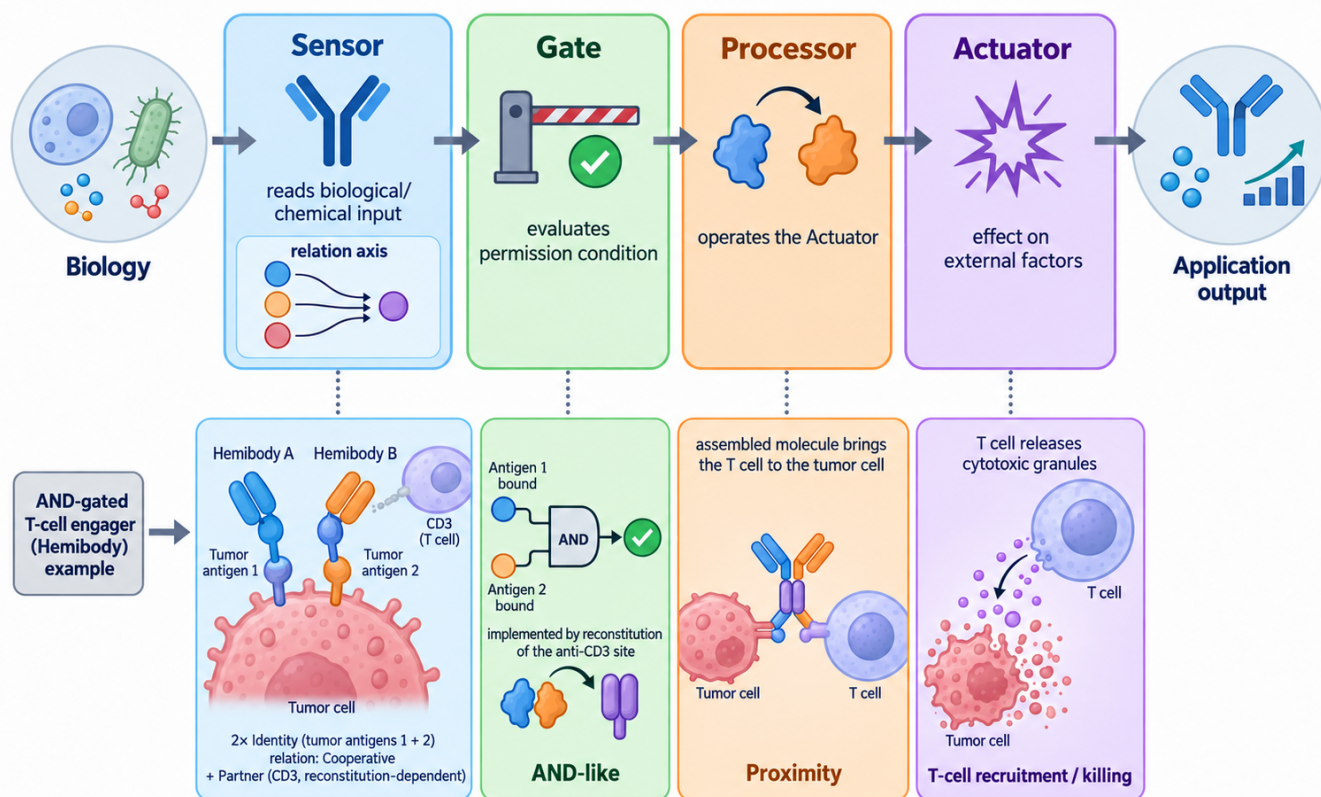


Figure 2: The Protein Engineering Grammar’s four primitives form a serial information-processing pipeline: Sensor, Gate, Processor, Actuator. Where an engineered protein carries more than one simultaneous Sensor instance, their outputs are first combined through an internal relation axis before being passed to the Gate. An AND-gated T-cell engager (hemibody type) instantiates this pipeline concretely (Sensor: two tumor-antigen Identity sensors combined through a Cooperative relation, plus a CD3 Partner sensor that becomes available only after the two half-antibodies assemble; Gate: AND-like, implemented by that reconstitution; Processor: Proximity; Actuator: T-cell recruitment and killing). Candidate classes for each primitive are enumerated in Figure 3.

remaining open to non-therapeutic instances as evidence permits (Section 3, below, includes one such example).

Figure 3 summarizes the full set of candidate classes for all four primitives, which we then define individually in the four sections below.

Sensor

A Sensor is the information that an engineered protein reads. We distinguish six candidate classes. **Identity** is recognition of a specific molecular entity, such as a target antigen or a peptide-MHC complex

[1]. **Context** is a condition or state of the surrounding environment: the activity of an enzyme such as a tumor-associated protease [9], the location of a molecule, or a physicochemical condition such as pH [10], oxygen level, temperature, or photon exposure [11]. **Partner** is recognition of another discrete molecule that participates in the protein’s mechanism, such as an Fc receptor [12] or a chemically induced dimerizer [13]. **Quantity** is a measurable density or concentration, such as antigen copy number per cell [14]. **Spatial** is active sensing of spatial context. **Self** is a time-dependent or stochastic change intrinsic to the molecule itself, such as the progress of spontaneous hydrolysis of a PEG-lysine conjugate [15]. The other five classes all reference an entity or variable

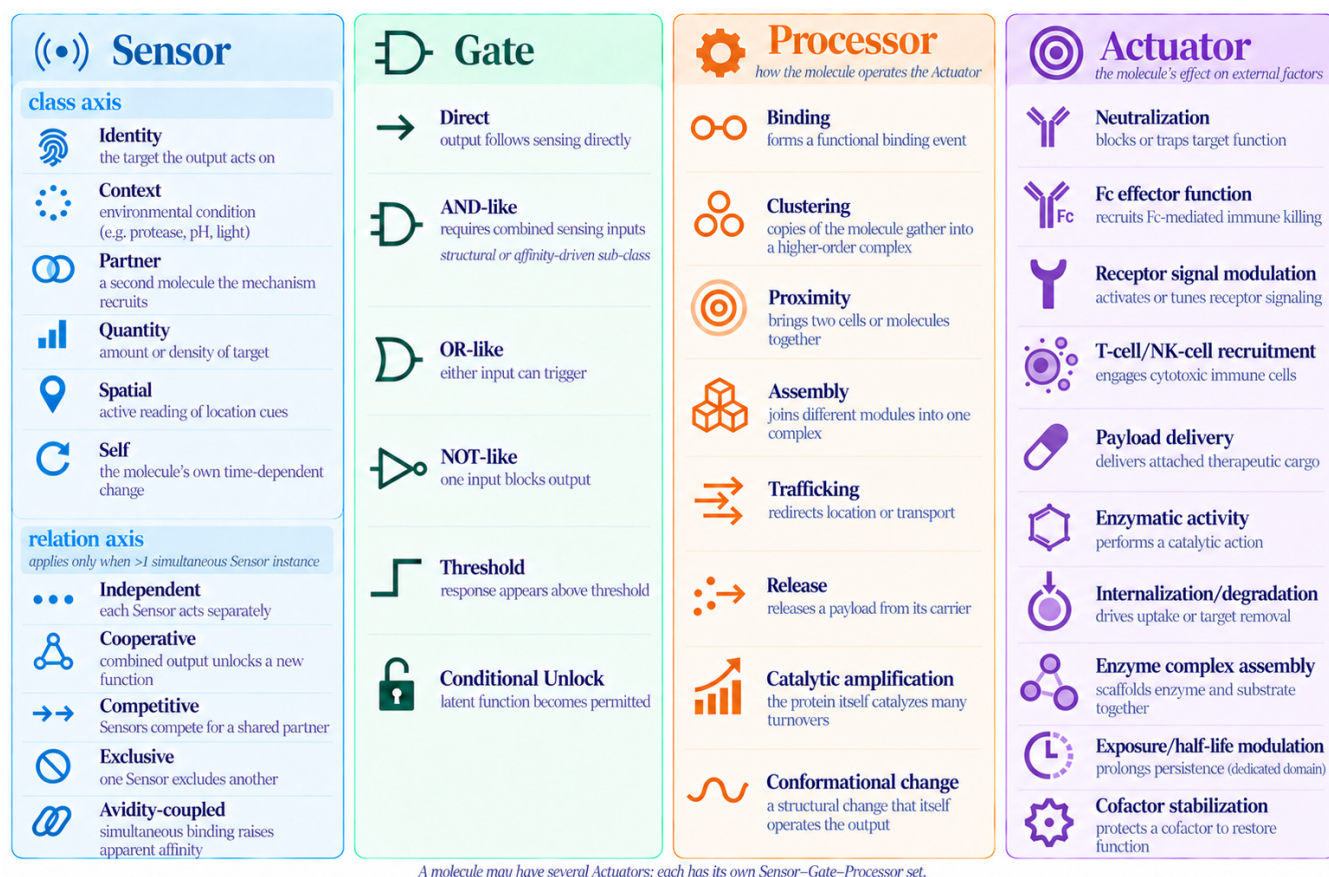


Figure 3: Candidate classes for each of the four Grammar primitives. Sensor: 6 classes along the class axis, plus 5 relation values combining multiple simultaneous Sensor instances. Gate: 6 classes (AND-like further divides into a structural and an affinity-driven sub-class). Processor: 8 classes. Actuator: 10 representative therapeutic-domain classes; not an exhaustive taxonomy.

external to the molecule. Self is distinguished by referencing none.

Three boundaries within this class merit explicit statement. First, Identity, Partner, and Context are distinguished by the sensed entity’s functional relationship to the Actuator, not by its chemical nature. Identity denotes the primary disease-relevant molecule that the Actuator itself acts upon — the target that is neutralized, agonized, or, in a cell-directed mechanism, the cell killed. Partner denotes a second, distinct molecule that is actively recruited or engaged to carry out the mechanism but is never itself the object of the Actuator’s action. Examples include an effector-cell receptor such as Fc R or CD3, a pharmacokinetics-relevant receptor such as FcRn, or a synthetic binding partner such as a chemically induced dimerizer. Context denotes a condition or state of the environment — an enzyme activity, a location, or a physicochemical condition such as pH or

light — that determines when or where the molecule should activate or localize, without itself being acted upon by the Actuator. A T-cell engager illustrates the Identity/Partner distinction directly: the tumor antigen is Identity, since T-cell-mediated killing is directed at the cell that bears it, while CD3 is Partner, recruited to engage a T cell but never itself the object of killing. A masked antibody illustrates Context: the activating protease is neither neutralized nor degraded by the antibody; its presence merely licenses the antibody to reach and act upon a separate Identity target. Second, Context covers discrete cues (the presence of a protease, an antigen at a location) and continuous physicochemical variables (pH, redox potential, light). We do not give them separate classes, because the line between them would depend on how a source describes the input rather than on what the molecule senses: protease activity, ATP, hypoxia, and pH all have a baseline level that changes in disease. A masked antibody that is activated by

tumor-associated protease activity senses Context [9], and so does an antibody engineered for pH-dependent antigen release [10]. Whether the variable is discrete or continuous is recorded in the notes of the mapping, not in the class. Third, Partner and Context are distinguished by whether the sensed input is itself a molecule. A chemically induced dimerization system senses a small-molecule ligand and is therefore a Partner Sensor, while a light-regulated allosteric switch senses photon exposure, which is not a molecule, and is therefore Context. This last criterion — is the input a molecule? — generalizes cleanly to sensing modalities outside chemistry and biology, including the light-based example discussed in Section 5.

When a protein carries more than one Sensor instance, we additionally record how their outputs are combined into a single analog signal prior to gating. There are five candidate values. Independent: Sensor outputs combine without cooperativity. Cooperative: the combined output unlocks or strengthens a downstream function that neither Sensor alone produces. Competitive: multiple Sensor instances compete for a shared resource or binding partner external to the molecule, an effect that can be graded and continuous. Exclusive: engagement of multiple Sensor instances is structurally mutually exclusive, through a shared binding site within a single molecule, an effect that is binary. Avidity-coupled: multivalent engagement whose apparent affinity depends on local target density. Competitive and Exclusive can be understood as two ends of the same continuum: Exclusive is the limiting case in which Competitive reaches complete structural constraint. Datasets nonetheless contain clear, independent examples at each end of that continuum, so we do not collapse them into a single value. Cooperative and Avidity-coupled, separately, are not two ends of a continuum. They are distinct axes answering different questions. Avidity-coupled answers a Sensor-layer question — does combined-instance engagement change the apparent strength of binding itself? Cooperative answers a functional-layer question — what does the combined output newly make possible downstream? The two are independently true or false, so a technology may have direct evidence for only one or for both. The five values are accordingly not a mutually exclusive single choice. These five values are distinguished by what happens when only one of the relevant binding events occurs.

A conventional IgG’s Fab (antigen recognition) and Fc (Fc R engagement) [12] illustrate Independent: in our reading, each engagement is structurally and functionally self-sufficient, and neither depends on or enhances the other (the cited structure documents the Fc-Fc RIII contact, not the independence itself). A dual-epitope agonist antibody pair engineered to hexamerize on the cell surface, in which each antibody binds a distinct, non-competing DR5 epitope [16], illustrates Cooperative: either antibody alone is functionally silent, and potent agonist signaling emerges only from their combined engagement. A de novo designed heterodimeric protein logic gate [4] illustrates Competitive: in its NOT-gate configuration, a designed input protein competitively binds a monomeric output protein that would otherwise homodimerize to reconstitute a split transcription factor, disrupting that homodimer and suppressing downstream gene expression — a mechanistically distinct route to the same relation value as the colocalization-gated decoy mechanism discussed under NOT-like below. A dual-specificity antibody that binds two structurally distinct antigens through a single, extensively overlapping paratope [17] illustrates Exclusive: because the two antigens’ contact surfaces overlap rather than being independently addressable, engaging one physically precludes engaging the other, a binary exclusion that contrasts with Competitive’s graded effect. A bivalent nanobody whose two domains recognize the same repeated epitope on von Willebrand factor [18] illustrates Avidity-coupled: both domains are genuine, simultaneous Sensor instances of the same Identity class. Their combination is neither Cooperative (neither domain’s binding enables a function the other lacks) nor Competitive or Exclusive (the domains engage simultaneously rather than excluding each other). Instead, it is a change in apparent binding strength that scales with how densely the target epitope is displayed. Avidity-coupled names this amplification, not the separate question of whether the combined instances address an identical target or genuinely distinct ones. A multivalent binder engaging several distinct epitopes at once could be avidity-coupled just as readily as one engaging repeated copies of a single epitope, and we treat the two axes (how many distinguishable targets are sensed; whether their combined engagement is affinity-amplifying) separately.

Gate

The Gate evaluates a permission condition on the Sensor’s output, combined across its relation axis where the Sensor comprises more than one instance. We use Boolean vocabulary — Direct, AND-like, OR-like, NOT-like, Threshold, Conditional Unlock — as an abstraction rather than a claim of strict digital logic. Biological responses are continuous, and we apply Boolean terms only where direct implementation evidence supports a threshold-like behavior. The internal ordering of a Processor’s steps (for example, binding followed by trafficking followed by payload release) is treated as a property of the Processor rather than of the Gate.

Each class denotes a distinct permission pattern. Direct denotes the absence of any explicit condition beyond the Sensor’s own output (including its relation axis, where applicable): in our reading, a conventional IgG’s Fc effector function (Fc R engagement as in [12]) proceeds once Fab and Fc engagement occur, with no further condition evaluated. A dual-epitope agonist antibody pair in which each antibody is silent alone [16] is also Direct, because it has no designed OFF state: productive receptor clustering needs both epitopes, and this conjunction is recorded by the Cooperative relation rather than by the Gate. AND-like requires several inputs, each independently needed to release a designed OFF state of the function, such as a mask, a latch, a split molecule, or a deliberately sub-threshold binding affinity. An engineered antibody pair designed to suppress homo-oligomerization while promoting hetero-oligomerization only upon co-engagement of two distinct antigens [19] produces negligible signaling unless both antigens are bound at once. How completely function is lost when one input is missing is not part of the definition. AND-like divides into two sub-classes by the kind of mechanism that implements it. In the structural sub-class, a change in the structure of the molecule releases the OFF state, for example masking, Fv reconstitution, or the formation of an Fc interface, including the Fc hetero-oligomerization switch above [19]. In the affinity-driven sub-class, a change in binding strength or avidity does, because the contributions of several arms add or multiply. Binding strength is inherently continuous, so function often diminishes gradually when an input is absent, as seen in a trispecific T-cell engager whose killing potency is dominated by its higher-affinity single arm, with only modest fur-

ther enhancement from engaging a second antigen [20]. Only the Sensors that release the OFF state are counted as inputs. The tumor antigen and CD3 of a conventional T-cell engager are not, because binding them is the function itself. OR-like permits function through any of several alternative inputs: a trispecific antibody combining three independent HIV-1-neutralizing epitopes [21] retains potent neutralization against viral strains that have escaped any single epitope, since a strain resistant to one arm remains sensitive to either of the other two. NOT-like suppresses an otherwise-permitted output in the presence of an additional input: a de novo designed cage-key protein switch activates upon colocalization on cells bearing two targeting antigens. In a three-antigen construct, however, output is actively suppressed when a third antigen’s targeting domain competitively sequesters the key away from the cage [22]. Threshold permits function only once a quantitative input crosses a defined level: a 2:1 T-cell-engaging bispecific’s activity tracks antigen density, with an empirical threshold of approximately 10,000 antigen-binding sites per cell separating high- from low-expressing cells [14], as measured across a panel of cell lines with differing antigen levels. Conditional Unlock denotes a permission condition whose satisfaction produces a lasting change in the molecule’s own accessibility rather than a reversible signal. A protease-masked antibody remains sterically blocked from its target until a tumor-associated protease cleaves the mask — an irreversible bond-cleavage event. After that, the antibody is free to engage its target [9]. A single input releases the OFF state; if several inputs are each independently needed, the Gate is AND-like. How a Conditional Unlock is implemented, for example by mask removal, allosteric change, induced association, or clustering, differs between technologies. We have not yet defined sub-classes of Conditional Unlock that would record these differences, and leave them for future work; AND-like already has two sub-classes.

Processor

The Processor is the molecule’s internal mechanism that operates the Actuator once a Gate permits it: binding, clustering, proximity-induced complex formation, assembly, trafficking, release, catalytic amplification, or conformational change. Where a

Processor comprises multiple ordered steps, that order is recorded as part of the Processor’s description. The Processor does not record how the Gate is implemented. Unmasking, reconstitution of a split molecule, competitive displacement, and avidity belong to the Gate’s implementation, and are recorded under the Gate’s sub-class. A T-cell engager, for example, has a Proximity Processor whether it is unmasked by a protease or assembled from two halves.

Each class denotes a distinct kind of physical event, several of which recur in ordered combination. Binding denotes a stoichiometric engagement event that is itself the entire physical consequence of a permitted Gate output, with no further rearrangement studied: a dual-specificity antibody engaging two structurally distinct antigens through a single, overlapping paratope [17] is recorded as Binding alone. Clustering denotes multiple copies of the molecule gathering into a higher-order complex, distinct from a single 1:1 binding event: a biparatopic antibody engaging two non-overlapping HER2 epitopes forms large complexes on the cell surface [23]. A single molecule that binds receptors and thereby brings them together, such as a cytokine or an agonist antibody acting alone, is recorded as Binding. Proximity-induced complex formation denotes the assembly of a stable multi-cell or multi-protein complex driven by simultaneous engagement of two separate entities: a bispecific T-cell engager (BiTE) induces a regular cytolytic immunological synapse between a T cell and a target cell, directly visualized by confocal microscopy for Ep-CAM $\times$ CD3 constructs [24], and the CD19 $\times$ CD3 BiTE blinatumomab shows clinical activity consistent with the same T-cell redirection [25]. Assembly denotes the joining of separately expressed or administered components into one product that is itself the output: a peptide tag forms a rapid covalent bond with its partner protein, and so joins any two proteins that carry them [26]. Reconstitution of a split molecule is not recorded as Assembly when it only implements a Gate. Trafficking denotes intracellular relocation following an initial binding or clustering event, most often toward a degradative or recycling compartment: for the biparatopic antibody above, HER2 clustering itself promotes internalization and lysosomal trafficking [23]. This illustrates how a single Processor description can record more than one ordered step: that same antibody, additionally conjugated to a cytotoxic payload, completes the chain with Release. Release denotes the liberation of a pay-

load from its carrier, as when a conjugated cytotoxic payload leaves an antibody-drug conjugate after lysosomal processing. Removal of a mask is not Release in this sense. A protease-masked antibody’s target-binding face remains sterically blocked until a tumor-associated protease cleaves the mask [9], but that cleavage implements the Conditional Unlock Gate, and the Processor is the Binding that follows. Catalytic amplification denotes multiple-turnover chemical transformation performed by the protein itself, rather than a single stoichiometric event. A monoclonal antibody raised against a transition-state analog of cocaine catalyzes repeated hydrolysis of the drug’s benzoyl ester bond at its own combining site, processing many substrate molecules per antibody rather than binding once [27]. Conformational change denotes a change in a molecule’s own three-dimensional structure that itself operates the Actuator, without changing how many separate modules or copies are involved. In a bispecific IgG with a rigid scaffold, occupancy of the HER2 arm constrains the orientation of the HER3 arm and so strengthens its blockade, a mechanism known as Dock and Block [28]. A structural change that only exposes a binding surface implements a Gate, and is not recorded here: a chemically switchable antibody exposes its antigen-binding surface only when a small-molecule-mediated allosteric mechanism, reinforced by an intrinsic pH-dependent component, shifts the paratope’s conformation [29], and its Processor is the Binding that follows. This class is distinct from Assembly or Clustering, in which the identity or number of interacting modules itself changes.

Actuator

The Actuator is the molecule’s effect on external factors: the final, domain-specific output. A molecule may have more than one Actuator. A T-cell engager whose tumor-antigen arm also neutralizes its target, for example, has two, T-cell recruitment and neutralization. Each Actuator has its own Sensor, Gate, and Processor assignment, which the tables label (A), (B), and so on, and every molecule has at least one Actuator. Within the therapeutic domain, representative classes include neutralization, Fc effector function, receptor signal modulation, T-cell or NK-cell recruitment, payload delivery, enzymatic activity, internalization or degradation, enzyme-complex assembly, cofactor stabilization, and exposure or half-life

modulation. Neutralization includes the antagonism and blockade of receptors and ligands, including cytokine trapping. Receptor signal modulation is the active generation or adjustment of a receptor’s signal, including that of cytokine receptors; it covers full agonism as well as partial or biased activation and control of the timing of a signal. Enzyme-complex assembly is a scaffolding function distinct from intrinsic catalysis, added to accommodate cofactor-mimetic bispecifics such as emicizumab. Cofactor stabilization covers a related but distinct case: a protein that performs no catalysis itself but binds and protects a non-protein cofactor from degradation, thereby indirectly restoring the function of a separate enzyme or cellular structure that depends on that cofactor. It differs from enzymatic activity in that the protein itself never catalyzes a reaction, and from enzyme-complex assembly in that one molecule is directly protected through binding rather than two molecules being brought into proximity. Exposure or half-life modulation was added to accommodate technologies whose entire function is to extend or tune a molecule’s persistence rather than to act on a target directly. This last class illustrates a general design point: not every engineered protein primarily produces a therapeutic effect on a target. Some primarily modulate the exposure of another effect instead, and the Grammar accommodates this as a distinct Actuator rather than folding it into an unrelated class. Exposure or half-life modulation counts as an Actuator of a molecule when a functional domain is placed in it mainly to extend its persistence, such as an anti-albumin domain or an Fc domain added for that purpose. It does not count when a physicochemical property of the molecule or domain improves the half-life, such as its size, PEGylation, or an introduced mutation. A technology whose only purpose is half-life extension is recorded with this Actuator, because it has no other. For non-therapeutic domains, gene expression control, biosensor signal output, and material or self-assembly output serve as placeholder classes.

The framework’s central claim and its scope

The generality of this vocabulary — its applicability beyond any single structural format — is the framework’s central claim. We do not propose that structural classifications such as “bispecific antibody” or “antibody-drug conjugate” are incorrect; rather,

we argue that they answer a different question than the one the Grammar answers. A structural classification describes what a molecule is built from. The Grammar describes what a molecule computes. Two molecules sharing a structural classification can implement entirely different logic (a point developed with concrete evidence-backed examples in Section 4). The reverse also holds, as the Introduction’s examples show: mechanistically similar logic can be implemented across structurally unrelated scaffolds, without requiring any extension to the four-primitive vocabulary described above.

4 Mapping Existing Modalities onto the Grammar

Dataset and evidence standard

We assembled a registry of protein engineering technologies (Table S1), spanning conventional IgG effector mechanisms, masked and conditionally activated antibodies, T-cell and NK-cell engagers, antibody-drug conjugates, de novo designed protein switches, and several deliberately non-antibody or non-therapeutic scaffolds. Primary literature was preferred over review or commentary sources, and company or marketing terminology was never used, on its own, to assign a Grammar class. Mechanistic assignments were made only where the underlying reference provided direct biochemical, cell-based, or in vivo evidence, and were otherwise left uncertain rather than inferred from the technology’s name or marketing description. Table S1 lists every registered technology with its Grammar assignment, an evidence status, a confidence level, and the primary references the assignment rests on.

All registered technologies fall within the scope this framework targets: a single engineered protein’s own Sensor-through-Actuator logic, evaluated independently of how that protein comes to be expressed.

Format and logic are separable

Table 1 visualizes the resulting technology-by-primitive matrix underlying the format/logic separation introduced in the Introduction and Section

Table 1: Grammar mapping of protein engineering technologies with a clearly identifiable disease-relevant molecular target and a Sensor/Gate/Processor/Actuator combination (including the Sensor relation value, which is not shown here) not already shown by another row, spanning therapeutic and non-therapeutic (mechanistically distinctive) examples (Table S1 in the supplementary dataset for full detail). Other registered technologies are excluded from this table: some because they either lack a single-molecule, disease-relevant target-directed mechanism (including platform/methodology technologies whose downstream application is not itself disease-specific) or because none of their primitive assignments are directly confirmed by the cited primary literature, others because they reproduce, on a different molecular target, a Sensor/Gate/Processor/Actuator combination already shown by another technology in this table, others because they only enhance a function of conventional IgG, and still others because each is a general class (for example, conventional IgG itself) of which specific molecules are already shown; all of them remain documented, with their full mapping, in the supplementary dataset. Each cell shows only a Sensor, Gate, Processor, or Actuator value that is supported by the cited primary literature or follows from the established mechanism of the technology; values that could not be assigned with confidence are left blank. Sensor cells give only the Sensor categories, without target details. When a molecule has more than one Actuator, the values that belong to each Actuator are labelled (A), (B), and so on, and are separated by semicolons; an unlabelled Sensor or Gate is shared by all Actuators.

Technology	Sensor	Gate	Processor	Actuator	Reference
Pembrolizumab	Identity	Direct	Binding	Neutralization	[30]
Teplizumab	Identity	Direct	Binding	Receptor signal modulation	[31], [32]
Trastuzumab	Identity	Direct	Binding → Trafficking → Release	Payload delivery	[33], [34]
Belantamab mafodotin	(A) Identity; (B) Identity + Partner	Direct	(A) Binding → Trafficking → Release; (B) Binding → Proximity	(A) Payload delivery/killing; (B) Fc effector function	[35]
Dupilumab	Identity	Direct	Binding	Neutralization	[36]
Anti-cocaine abzyme	Identity	Direct	Catalytic amplification	Enzymatic activity	[27]
synNotch	Identity	Conditional Unlock	Trafficking	Gene expression control	[5]
Conditionally-activated cytosol-penetrating antibody	Context	Conditional Unlock	Trafficking	Payload delivery	[37]
Orthogonal IL-2/IL-2R pair	Partner	Direct	Binding	Receptor signal modulation	[38]
Effector-triggered facilitated-dissociation kinetic switch	Partner	Direct	Binding → Conformational change	Receptor signal modulation	[39]
Velocinator	Partner	Direct	Binding → Trafficking	Enzymatic activity	[40], [41]
ABD half-life extension	Partner	Direct	Binding	Exposure / half-life modulation	[42]
LITE switch	Partner	Conditional Unlock	(A) Binding; (B) Binding; (C) Proximity	(A) tumor localization of a radionuclide-conjugated antibody; (B) Exposure / half-life modulation; (C) T-cell recruitment	[43]
LOCKR	Partner	Conditional Unlock	(A), (B), (C) Binding	(A) partner binding by the motif on the latch; (B) degradation of the fusion; (C) nuclear export. Classes for (A)-(C) are not in the Grammar list	[22], [44]
Optogenetic allosteric switch	Context	Direct	Conformational change	Enzymatic activity	[11]
PaCS anti-CTLA4 switch antibody	Context	Conditional Unlock	Binding	Neutralization	[29]
Lonapegsomatropin	Self	Direct	Release	Receptor signal modulation	[45]
Biparatopic ADC	2x Identity		Clustering → Trafficking → Release	Payload delivery	[23], [46]
Two-in-One antibody	2x Identity	Direct	Binding	Neutralization	[17]
MEDI5752	2x Identity	Direct	Binding + Internalization/degradation	Neutralization	[47]
ALX-0081	2x Identity	Direct	Binding	Neutralization	[18]
Faricimab	2x Identity	Direct	Binding	Neutralization	[48], [49]

Table 1: (continued)

Technology	Sensor	Gate	Processor	Actuator	Reference
Zenocutuzumab	2x Identity	Direct	Binding → Conformational change	Neutralization	[28]
Rilonacept	2x Identity	Direct	Binding	Neutralization	[50], [51]
ZW1528	2x Identity	Direct	Binding	Neutralization	[52]
Davutamig	2x Identity	Direct	Clustering	Internalization / degradation	[53], [54]
Zanidatamab	2x Identity	Direct	(A) Clustering; Proximity; (B) Binding; (C) Clustering → Trafficking	(A) Fc effector function; (B) Neutralization; (C) Internalization/degradation	[55]
Emicizumab	2x Identity/Partner	Direct	Proximity	Enzyme complex assembly	[56], [57]
Mim8	2x Identity/Partner	Direct	Proximity	Enzyme complex assembly / catalytic scaffolding	[58]
HexElect	2x Identity	AND-like	(A), (B) Clustering	(A) Fc effector function; (B) Receptor signal modulation	[19]
HexaBody-DR5	2x Identity	Direct	Clustering	Receptor signal modulation	[16]
Co-LOCKR	Multiple Identity	AND-like / NOT-like / OR-like	Binding → Proximity	T-cell recruitment	[22]
Hemibody	2x Identity	AND-like	Proximity	T-cell recruitment/killing	[59]
Bimekizumab	2x Identity	OR-like	Binding	Neutralization	[60], [61]
Bicycle toxin conjugate	Identity + Context	Direct	Binding → Trafficking → Release	Payload delivery	[2], [62]
Switch-Ig	Context + Identity	Conditional Unlock		Receptor signal modulation	[63], [64]
FAP-4-1BBL	Context + Identity	Conditional Unlock	Binding	Receptor signal modulation	[65], [66], [67]
Probody	Context + Identity	Conditional Unlock	Binding	Neutralization	[9], [68]
GlyTR	Identity + Partner			T-cell recruitment/killing	[69]
Nirsevimab	Identity	Direct	Binding	Neutralization	[70]
Trastuzumab	(A) Identity; (B) Identity + Partner	Direct	(A) Binding; (B) Binding → Proximity	(A) Neutralization; (B) Fc effector function	[71]
ADG106	(A) Identity + Partner; (B) Identity	Direct	(A) Binding → Clustering; (B) Binding	(A) Receptor signal modulation; (B) Neutralization	[72]
Boltbody	Identity + Partner	Direct	Binding → Trafficking → Release	Payload delivery	[73]
Obinutuzumab	(A) Identity; (B) Identity + Partner	Direct	(A) Binding; (B) Binding → Proximity	(A) Receptor signal modulation; (B) Fc effector function	[74], [75], [76]
Rituximab	Identity + Partner	Direct	Proximity	Fc effector function	[77]
Blinatumomab	Identity + Partner	Direct	Proximity	T-cell recruitment	[24], [25]
ImmTAC	Identity + Partner	Direct	Proximity	T-cell recruitment/killing	[1], [78]
AFM13	Identity + Partner	Direct	Proximity	NK-cell recruitment	[79]
AB248	Identity + Partner	Direct	Proximity	Receptor signal modulation	[80]
AbTAC	Identity + Partner	Direct	Proximity	Internalization / degradation	[81]

Table 1: (continued)

Technology	Sensor	Gate	Processor	Actuator	Reference
Immuno-STAT / Neo-STAT	Identity + Partner	AND-like	Binding	Receptor signal modulation	[82]
Efgartigimod	Identity + Context	Direct	Binding $\rightarrow$ Trafficking	Neutralization	[83]
DNL593	Identity + Context	Conditional Unlock	Binding	Cofactor stabilization	[84]
L19-IL2	Context + Partner	Direct	Binding	Receptor signal modulation	[85], [86]
Protease-activated cytokine prodrug	Context + Partner	Conditional Unlock	Binding	Receptor signal modulation	[87]
Trispecific anti-HIV-1 bnAb	3x Identity	OR-like	Binding	Neutralization	[21]
Amivantamab	(A) 2x Identity; (B) 2x Identity + Partner	Direct	(A) Binding; (B) Binding $\rightarrow$ Proximity	(A) Neutralization; (B) Internalization / degradation	[88], [89]
Glofitamab	2x Identity + Partner	Direct	Proximity	T-cell recruitment	[90]
Xaluritamig	2x Identity + Partner	Direct	Proximity	T-cell recruitment/killing	[91], [92]
AMG 305	2x Identity + Partner	AND-like	Proximity	T-cell recruitment/killing	[93], [94]
Anti-HER2 x anti-CD3 T-cell-dependent bispecific XPAT	2x Identity + Partner	Threshold	Proximity	T-cell recruitment/killing	[95]
ADG206	Context + Identity + Partner	AND-like	Proximity	T-cell recruitment/killing	[96]
ADG138	Context + Identity + Partner	Conditional Unlock	Binding $\rightarrow$ Clustering	Receptor signal modulation	[97]
Tarlatamab	(A) Identity + Partner; (B) Partner	AND-like	Proximity	T-cell recruitment/killing	[98]
TriTAC	(A) Identity + Partner; (B) Partner	Direct	(A) Binding $\rightarrow$ Proximity; (B) Binding $\rightarrow$ Trafficking	(A) T-cell recruitment/killing; (B) Exposure / half-life modulation	[99], [100]
ZW209	(A) Identity + Partner; (B) Identity + Partner; relation between and: Exclusive	Direct	(A) Binding $\rightarrow$ Proximity; (B) Binding $\rightarrow$ Trafficking	(A) T-cell recruitment/killing; (B) Exposure / half-life modulation	[101], [102]
Dual-Ig	Identity + 2x Partner	Direct	Binding $\rightarrow$ Proximity	T-cell recruitment/killing	[103]
Cibisatamab	(A) Identity + Partner; (B) Identity + Partner; relation between and: Exclusive	Direct	(A), (B) Proximity	(A) T-cell recruitment/killing; (B) Receptor signal modulation	[104]
Crovalimab	Identity + Partner + Quantity	Threshold	Proximity	T-cell recruitment/killing	[14], [105], [106]
acid-Fc	Identity + Partner + Context	Direct	Binding $\rightarrow$ Trafficking $\rightarrow$ Release	Neutralization	[107], [108]
Sotatercept	Context + Partner + Identity	Conditional Unlock	Proximity	Fc effector function	[109]
Restoret	4x Identity	Direct	(A), (B) Binding	(A) Neutralization; (B) Receptor signal modulation	[110]
ProTriTAC	(A) Context + Identity + Partner; (B) Partner	Direct	Binding	Receptor signal modulation	[111]
JANX008	(A) Context + Identity + Partner; (B) Partner	(A) Conditional Unlock; (B) NOT-like	(A) Proximity; (B) Binding	(A) T-cell recruitment/killing; (B) Exposure / half-life modulation	[112], [113]
IPH6501	(A) Identity + 2x Partner; (B) Partner	(A) AND-like; (B) NOT-like	(A) Proximity; (B) Binding	(A) T-cell recruitment/killing; (B) Exposure / half-life modulation	[114]
Ensovibep	(A) Identity + 2x Partner; (B) Partner	(A) Identity + 2x Partner; (B) Partner	(A) Identity + 2x Partner; (B) Partner	(A) NK-cell recruitment/killing; (B) Receptor signal modulation	[115]
	(A) 3x Identity; (B) 2x Partner	Direct	(A) Binding; (B) Binding	(A) Neutralization; (B) Exposure / half-life modulation	[116]

3: molecules sharing an identical structural classification can implement distinct functional logic, and, conversely, molecules built from unrelated structural scaffolds can implement equivalent logic.

Recurring patterns

One pattern recurred often enough across the dataset to warrant explicit note. Fc engineering does not concentrate in a single primitive. It recurs as a lever across nearly every layer of the Grammar. Fc modification is used to tune a Partner Sensor’s affinity for an effector receptor, as in afucosylation enhancing Fc RIIIa engagement. It is used to gate an Assembly event, as in an engineered Fc interface that suppresses homo-oligomerization while promoting hetero-oligomerization only upon co-antigen engagement. And it is used to determine overall Protein Architecture, as in Fc silencing to eliminate an unwanted effector pathway, or monovalent formats. Rather than treating “Fc engineering” as a single technology category, we found it more informative to classify each instance by which Grammar primitive it targets, since a shared physical substrate (the Fc domain) is put to mechanistically distinct uses across different technologies.

Absent combinations (white-space)

What follows is an inference about absent combinations of primitives, identified through a targeted counterevidence search against that same evidence base. A classification framework that only organizes known examples has limited value. A more demanding test is whether it can point toward combinations of primitives that existing modalities have not yet explored. We generated and evaluated candidate white-space opportunities from gaps observed while carrying out the Grammar mapping above. One of them, disease classifier discovery, is addressed in Section 5 (under Domain Classifier), since it is not a question about a combination of Grammar primitives but a qualitatively different one concerning the Compiler’s Domain Classifier stage itself. This section evaluates the remaining six against three independent signals. First, we assess how far each opportunity is already represented among our evidence-backed technologies. Second, we examine what a targeted primary-literature search for counterevidence outside

that dataset finds. Third, we weigh whether the corresponding technology has reached approval or market stage, together with the size of the remaining clinical need. This is consistent with the review posture we apply throughout this work (Table 2). We give equal focus below not only to areas with no implementation at all, but to areas with proof of principle that have not yet reached the market as a therapeutic application, and to areas with few implementing technologies but high remaining clinical need.

Temporal or kinetic logic

A Gate whose permission condition depends on the relative timing or sequence of two or more independent inputs — rather than on their simple co-presence — has no implementation in our dataset. We have previously assigned ordered multi-step Processors (for example, binding followed by trafficking followed by payload release in an antibody-drug conjugate) to the Processor primitive rather than to a distinct temporal Gate. Two partial precedents come closest to this space: a time-dependent activity change driven by a single intrinsic PEG-hydrolysis timer (bempegaldesleukin) [15], and an effector-triggered dissociation-rate control (up to 5,700-fold) that provides external, second-timescale silencing [39]. Neither, however, satisfies Gate logic conditioned on the relative timing of two or more independent signals. This absence is not merely a technical gap: there is a real clinical need — cytokine therapies and T-cell engager therapies require temporal control to suppress toxicity (such as cytokine release syndrome) while preserving efficacy. We report this as the strongest candidate satisfying all three conditions simultaneously: no implementing technology, no market presence, and high clinical need. Our literature search confirms that this covers the cases we are aware of: published work on protein switch kinetics addresses the engineering of a single switch’s response rate [39], not a Gate keyed to the interval or order between two independent external signals.

Multi-context sensing and true NOT-like Gate: a translational gap

Multi-context sensing and true NOT-like Gate are two candidates that share the same kind of gap: both

Table 2: Six candidate white-space opportunities evaluated in this section. A seventh candidate, Disease Classifier Discovery, bears on the Compiler’s Domain Classifier stage itself and is reported in Section 5 rather than here.

Candidate	Importance	Feasibility	Status
Temporal / Kinetic Logic	Medium-High	Low-Medium	No implementation found; top-ranked candidate
Competitive / Ratiometric Sensing	Medium	Medium-High	No implementation found; translational gap from an established biosensor principle
Multi-context Sensing	Medium	Medium	Demonstrated in principle (platform-stage); no disease-specific application yet
True NOT-like Gate	Medium	Medium	Demonstrated in principle (platform-stage); revised downward from an initial top-ranked hypothesis after two independent counterexamples
Geometry-to-Function	High	Medium	Partial precedent; substantially, though not completely, explored
Tunable Threshold	Medium	Medium	Underlying antigen-density/avidity gating already substantially explored; the externally tunable (post-administration) variant remains an open sub-question

are demonstrated in principle within our dataset, but neither has advanced beyond preclinical proof of principle toward a therapeutic application with a disease target.

Multi-context sensing is a mechanism in which a Gate integrates multiple independent, discrete environmental cues, rather than multiple copies of the same cue type. In principle, it is demonstrated by a platform that gates three orthogonal proteases AND-like or OR-like through molecular topology [117]. That literature, however, names no specific disease target or clinical candidate. It remains a platform/methodology-stage demonstration. The same is true of a true NOT-like Gate — a mechanism that is constitutively active by default and is actively suppressed, rather than activated, by the presence of a specific input. A de novo designed, colocalization-dependent protein switch [22] demonstrated an explicit, primary-literature-evidenced NOT-like Gate: a three-input construct adds a targeting domain that competitively sequesters the key, and requires no host-cell genetic modification at all. This overturned an initial structural-asymmetry argument (that AND/OR logic follows naturally from multivalent binding while NOT-like suppression does not), but this demonstration, too, remains at the platform/methodology stage rather than an approved or clinically advanced molecule. A de novo logic-gate platform [4] provides a second, mechanistically independent example, implementing a NOT-like Gate through the competitive dimer disruption of two designed heterodimeric proteins. This one, however, was demonstrated in yeast and primary human T cells, where it requires host-cell genetic modification,

and its cellular output is a generic transcriptional readout rather than a disease-specific one. Together, these examples show that the NOT-like Gate mechanism is reachable from more than one independent design strategy rather than being a single isolated implementation — but the conclusion of a translational gap to therapeutic application is unchanged.

Both mechanisms directly resonate with a direction strongly sought across the bispecific antibody and cell therapy pipeline as a whole: reducing on-target/off-tumor toxicity by gating on a combination of biomarkers, or on the presence of a third, normal-tissue-marker antigen. We report these not as complete white space in principle, but as regions that remain worth focused attention because translation to a therapeutic application has not yet been achieved.

Competitive or ratiometric sensing

A Gate that responds continuously to the ratio of two circulating markers’ concentrations rather than to either one’s absolute presence represents a different kind of gap. The underlying sensing principle is well established in the biosensor literature — dichromatic, ratiometric fluorescence readouts are a standard technique for concentration-independent measurement [118] — but we found no example of this principle being used to drive a therapeutic Actuator rather than a diagnostic readout. The closest counterpart in our dataset is the 2:1 T-cell bispecific’s empirically observed antigen-density threshold (approximately 10,000 antigen-binding sites per cell; Section 3), a fixed absolute value that does not adapt to a given

patient’s baseline biomarker level. Ratio-based activation would compensate for this rigidity and could improve robustness to inter-patient variability. We treat this as a translational gap rather than a technical unknown.

The closest candidate was a winner-take-all comparator circuit that compares the relative levels of two co-expressed proteins and uses the result to control cell death [119] — a mechanism structurally close to ratiometric sensing. We did not add it to our registry: the circuit spans up to eight co-expressed protein species, exceeding the single-molecule premise this framework treats as out of scope (Section 2), and its inputs are the expression levels of transfected synthetic constructs rather than circulating biomarkers. We record this as a boundary observation rather than a counterexample. Our assessment that ratiometric sensing has not yet been implemented as a therapeutic Gate is unchanged.

Other candidates

The remaining two candidates each fail to satisfy at least one of our evaluation axes — the number of implementing technologies, market status, or urgency of need — and we report them briefly. A tunable threshold — a Gate’s activation cutoff adjustable externally after administration, rather than fixed at the design stage — remains unconfirmed in our dataset, even though antigen-density/avidity gating itself is already heavily represented, by an explicit, quantitatively threshold-gated T-cell engager and by several cooperative-binding technologies; we note the tunable variant only as a narrower open question. Geometry-to-function (a mechanism in which the spatial arrangement of binding domains, rather than their chemical identity, determines the functional outcome) is partially satisfied by a T-cell engager [120] that ties the immune synapse’s intermembrane distance, as a continuous parameter, to functional magnitude. Realizing a specified geometry is, with current AI-driven structure design methods (Section 6), no longer the binding constraint; the genuinely open problem is hypothesis generation — determining which geometry a given function actually requires in the first place.

5 The Protein Engineering Compiler

From target selection to disease classifiers

Conventional therapeutic protein design is often framed as a three-step process: identify a disease, identify a target, design a drug against that target. This framing treats target selection and molecular design as largely independent decisions. We propose an alternative framing, in which the transformation from a biological problem to an engineered protein is made explicit as a seven-stage pipeline general enough to apply across application domains — the field in which the pipeline is instantiated, such as therapeutics, industrial enzyme engineering, or a non-therapeutic research tool (a worked example of the latter appears later in this section). The seven stages are: Problem Domain Biology, Feature Selection, Domain Classifier, Molecular Implementation, Protein Architecture, Constraint Layer, and Engineering Output. Three of these seven stages — the first, third, and last — carry a domain-specific name. The remaining four (Feature Selection, Molecular Implementation, Protein Architecture, and Constraint Layer) keep the same name and role regardless of which domain the pipeline is applied to. In therapeutics, the domain this manuscript focuses on throughout, the three domain-specific stages are named Disease Biology (in place of Problem Domain Biology), Disease Classifier (in place of Domain Classifier), and Therapeutic Output (in place of Engineering Output) (Figure 4).

Each stage has a distinct role. Problem Domain Biology and Engineering Output are domain-specific bookends. We take the remaining five stages in turn.

Feature Selection

Feature Selection identifies which measurable biological or chemical variables an engineered protein could plausibly sense from the relevant biology. These variables correspond to the terms $S_1, \dots, S_n$ in the classifier formalism below.

Domain Classifier

The Domain Classifier formalizes the selected Feature variables into a decision function, $\text{Classifier} = f(S_1,$

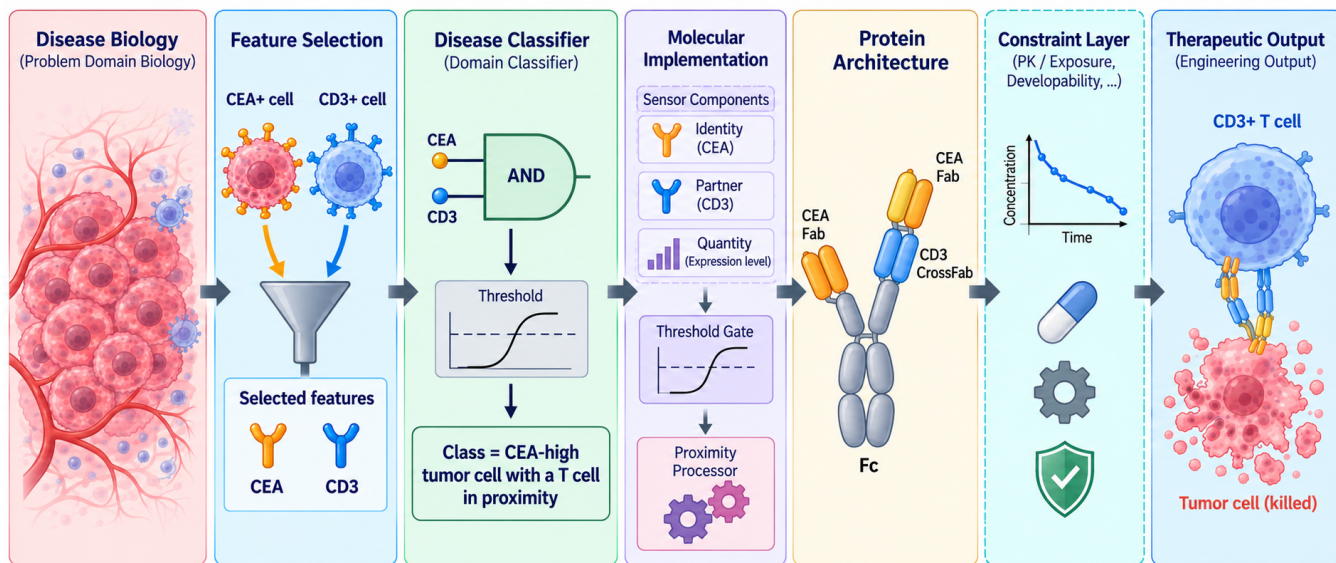


Figure 4: The Protein Engineering Compiler formalizes the transformation from a biological problem to an engineered molecular output as a seven-stage pipeline: Problem Domain Biology, Feature Selection, Domain Classifier, Molecular Implementation, Protein Architecture, Constraint Layer, and Engineering Output. The therapeutic domain instantiates this general form as Disease Biology through Therapeutic Output.

$S_2, \dots, S_n$), that distinguishes the biological state of interest (for example, a diseased cell) from the states it must be distinguished from. The logical pattern this function implements — for example, “both feature A and feature B must be present,” or a threshold on a single feature — is part of this formalization, stated independently of any specific molecular mechanism.

This stage itself leaves an open question. In our dataset, every Domain Classifier is built on a combination of biological features that an investigator selected in advance. Disease classifier discovery is different: rather than acting on a combination chosen in advance by the investigator, a system would itself identify which combination of features best distinguishes diseased from healthy tissue. We find no implementation of this. It is not a question about a combination of Grammar primitives. It is, instead, a question one stage further upstream than Molecular Implementation, about how the Domain Classifier’s logical pattern itself is constructed. In vitro selection technologies discover binders against a pre-specified target, not new combinations of features. We are not aware of a system that discovers new feature combinations at the molecular level. This bears directly on what the Compiler is ultimately meant to deliver,

which is why we judge it the most consequential opportunity in principle. The task of a system identifying feature combinations for itself is, however, already well advanced in the context of omics research. What we are pointing to is the absence of a system that does this at the molecular level. We therefore report this not as a long-term research direction for protein engineering alone, but as a long-term prospect at the intersection of omics-based classifier discovery and protein engineering, two currently distinct research areas.

Molecular Implementation

Molecular Implementation is where the Grammar itself enters the pipeline: it is the choice of which Grammar primitive class (Sensor through Actuator; Section 3) will realize that logical pattern. Critically, the same Domain Classifier can often be satisfied by more than one Molecular Implementation.

Protein Architecture

Protein Architecture is the specific physical scaffold used to build the chosen Molecular Implementation —

isotype, valency, domain arrangement, chain composition, conjugation chemistry. The same Molecular Implementation can, in turn, be realized through more than one Protein Architecture (an AND-like Gate, for example, need not be implemented through Fc heterooligomerization specifically). Purely manufacturing-time chemistry, such as a dual-payload antibody-drug conjugate’s site-specific dual-cysteine conjugation, is Protein Architecture and nothing else. There is no Sensor-through-Actuator description that duplicates it and blurs the boundary with Protein Architecture. This layer can exist entirely without any mechanism information: papers describing manufacturing or conjugation chemistry, or a de novo structure design method in itself, rarely fill in the Grammar fields. The reverse case also occurs — Fc engineering, for example, where a single physical substrate crosses several Grammar layers (Partner Sensor tuning, Assembly Gate control, Protein Architecture constraint) depending on the technology.

Constraint Layer

An engineered protein must also satisfy constraints that do not belong to the Sensor-through-Actuator chain itself. Most centrally, these are pharmacokinetics (PK) and exposure (half-life, clearance, tissue distribution), and developability (manufacturing suitability, such as thermal stability and aggregation propensity). We place these in a Constraint Layer positioned between Protein Architecture and Engineering Output. This is not a fixed list of phenomena but a relative classification: the same receptor engagement can be Grammar-level in one technology and Constraint Layer in another, depending on whether that engagement affects the function currently being analyzed. FcRn-mediated recycling illustrates this. Where recycling is itself a technology’s designed focus, it enters the Grammar as a Sensor (an engineered pH-dependent recycling antibody derived from tocilizumab, whose pH dependence is a Context Sensor [10]). Where it merely modulates the exposure of a separately targeted therapeutic effect, it is bracketed into the Constraint Layer (a transferrin-receptor-engineered bispecific antibody that promotes blood-brain-barrier transcytosis [121]).

Formal abstraction and worked examples

We formalize the Domain Classifier as $\text{Classifier} = f(S_1, S_2, \dots, S_n)$, and the resulting engineered protein as the product of that classifier with a Processor and an Actuator: $\text{Engineered Protein} = \text{Classifier} \times \text{Processor} \times \text{Actuator}$. The Classifier evaluates to 1 when its permission condition is satisfied and 0 otherwise, so the Processor-Actuator product — the engineered protein’s function — proceeds only when the Classifier equals 1. This formalism is not merely notational. Worked examples grounded in evidence-backed technologies from our dataset instantiate it concretely across different Gate types. A 2:1 T-cell bispecific [14] instantiates a threshold classifier — $\text{Classifier}(\text{cell}) = 1$ if antigen density exceeds an explicit, quantified threshold and a CD3-positive T cell is proximate — with an empirically observed numeric threshold (approximately 10,000 antigen-binding sites per cell). A protease-activated cytokine prodrug [87] instantiates a binary presence classifier gated on local tumor-associated protease activity, with a quantitatively confirmed on/off differential (the intact prodrug binds the IL-2 receptor with 13-fold lower affinity and induces 8.5-fold less signaling than its unmasked counterpart). A logic-gated antibody pair [19] instantiates an AND-like classifier requiring co-expression of two distinct antigens on the same cell, with single-antibody controls confirming that neither antigen alone triggers the effector mechanism. A trispecific broadly neutralizing antibody [21] instantiates an OR-like classifier over three independent viral epitopes, with strain-specific single-arm sensitivity and resistance patterns confirming that the three arms act independently rather than requiring simultaneous engagement. This case also demonstrates the Grammar’s Gate vocabulary generalizing cleanly from two to three simultaneous Sensor inputs.

A non-therapeutic instance

A further worked example departs from the therapeutic domain entirely. A light-regulated allosteric switch module [11], used as a research tool to control kinase and recombinase activity in cultured cells, instantiates a classifier over photon exposure rather than any biological feature: it evaluates to 1 when

light matching the engineered chromophore’s activation spectrum is present, and 0 otherwise. We include this example for two reasons. First, it directly answers a question our own Compiler specification had left open — what does a Domain Classifier look like in a domain other than disease biology. Second, in this instance, Feature Selection operates over an input (light) that is not a biological feature at all, demonstrating that the general form of the Compiler abstraction is not intrinsically tied to biological sensing.

6 Toward Programmable Molecular Therapeutics

Grammar and Compiler as a harness for AI-driven engineering

The Grammar (Sensor-Gate-Processor-Actuator) and Compiler (Problem Domain Biology through Engineering Output) organized in this paper are, at present, a descriptive and diagnostic vocabulary rather than a generative tool that produces novel molecules (Section 7). We suggest, however, that this same decomposition may serve a distinct purpose for AI-driven protein engineering. Rather than treating design as a single end-to-end black box, it structures the problem as a chain of intermediate representations that can be inspected, modified, and evaluated independently at each stage — the kind of structure a harness provides.

Individual steps in current generative protein design pipelines — hallucination-based backbone generation, inverse folding, structure prediction, and gradient-based multi-objective optimization, alone or in combination — are already highly automated. What remains dependent on human domain expertise is the layer that connects these steps to a higher-level design intent: which Grammar primitive a given step is meant to realize, and which Domain Classifier logic pattern that primitive is meant to satisfy. The intermediate representations this paper’s Compiler formalism makes explicit — Feature Selection, Domain Classifier, Molecular Implementation, Protein Architecture — structure that connection as a chain of separately checkable sub-problems rather than a single undifferentiated optimization. In this sense, the vocabulary proposed here may function as a set

of checkpoints at which a human can inspect and correct the output of an AI system carrying out end-to-end design — a harness for AI-driven engineering in that specific sense.

The underdeveloped state of Problem Domain Biology identification

Of the Compiler’s seven stages, we have treated Molecular Implementation and Protein Architecture in the greatest detail (Section 5), because our evidence base is grounded in mechanistic literature describing those stages. By contrast, the pipeline’s first stage, Problem Domain Biology (Disease Biology, in the therapeutic instance), is treated in this paper as a single node whose internal structure we have not formalized.

There is little reason to expect that stage to be simple. Deciding which biological feature an engineered protein should sense (Feature Selection) presupposes an earlier process: identifying and prioritizing, from an effectively unbounded set of candidate features — gene expression, protein-protein interactions, metabolite concentrations, cell-surface markers, and others — those that best discriminate the disease state of interest. This is likely to require the same kind of multi-step judgment we have described for Molecular Implementation and Protein Architecture — enumerating candidates, narrowing them against evidence, weighing trade-offs among competing feature combinations — rather than resolving at a single node. It plausibly has a detailed, compiler-like structure of its own.

The white space we reported in Section 5 as disease classifier discovery is the other side of the same underdevelopment we describe here. Rather than acting on a combination of features an investigator chose in advance, the same system would itself discover which target, and which combination of features, to select in the first place.

This gap matters specifically for any attempt to extend AI-driven engineering across the full pipeline rather than to the stages from Molecular Implementation onward. Generative methods for the Molecular Implementation-to-Protein Architecture transition have advanced rapidly over the past decade, but these methods take the Domain Classifier as a given input. They do not address the upstream problem of

constructing that input in the first place. Running the full protein engineering pipeline in an AI-driven manner will, in our view, require formalizing the process from Problem Domain Biology through Feature Selection to Domain Classifier in comparable detail to what has already been achieved downstream of it. We flag this as a direction for future work rather than a problem this paper resolves.

7 Conclusion

We have presented a vocabulary for describing engineered proteins by the information-processing logic they implement rather than by the structural format they are built from. We evaluated it against a broad, evidence-graded set of real technologies, and used it to surface both confirmed and refuted hypotheses about which regions of the resulting design space remain unexplored. The framework’s value at this stage is organizational and diagnostic. It makes explicit that structural classification and functional logic are separable axes, and it provides a shared vocabulary for discussing molecular mechanism across otherwise unrelated engineering platforms. Whether this vocabulary can additionally support generative design — proposing and validating novel molecules rather than classifying existing ones — is the central open question this work leaves for future evaluation.

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